LESSON 4.1

Understand Percent

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can explain a percent as a rate per 100, model it on a decimal grid or tape diagram, and interpret percents greater than 100%.

WHY THIS MATTERS The arcade posts every result as a percent of the weekly goal. A percent compares a number to 100 — so 40% means 40 out of every 100. Some players land under the goal, and some blow past it: a score of 150% means one and a half times the whole goal. Your job is to read and model each result as a comparison to 100.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Percent *Spanish: Porcentaje*	A ratio that compares a number to 100. The word means “per hundred,” and the symbol is %.	40% = 40 out of every 100
Per hundred *Spanish: Por cada cien*	The meaning hidden inside the word percent: for every group of 100. It is why every percent can be written as a fraction over 100.	7% means 7 per hundred
Decimal grid *Spanish: Cuadrícula decimal*	A 10-by-10 grid of 100 squares. Shading squares shows a percent directly, one square per percent.	62 shaded squares = 62%
Part-to-whole *Spanish: Parte a todo*	A comparison of one part against the whole amount. In a percent, the whole is always 100%.	shaded squares to all squares
Greater than 100% *Spanish: Mayor que 100 %*	More than one whole. A percent above 100 means the amount is larger than the thing it is being compared to.	150% of the gray kitten = 1.5 times its weight
Equivalent ratio *Spanish: Razón equivalente*	A ratio with the same value, made by multiplying or dividing both terms by the same number.	2 : 5 → 40 : 100

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

3 out of 4 students bring lunch. What percent?

1. What times 4 makes 100? 25.
2. 3 × 25 = 75 and 4 × 25 = 100.
3. 75 out of 100 = 75%.

Answer: 75% bring lunch

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Shanice's gray kitten weighs 4 ounces. The orange kitten weighs 1.5 times as much. What percent of the gray kitten's weight is the orange kitten's weight?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- _____ % means _____ out of every hundred.
- To compare to a hundred, I multiplied both terms by _____ .
- The part is _____ and the whole is _____ .
- A percent is a ratio because _____ .

Word bank percent | per hundred | ratio | part | whole | equivalent | multiply | terms | grid | compare

WATCH OUT!

A common mistake

A common mistake with percent is scaling only one term of the ratio: reading 7 out of 10 as 7%. A percent compares to 100, so BOTH terms must be multiplied by the same factor

LESSON 4.1

Understand Percent

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can explain a percent as a rate per 100, model it on a decimal grid or tape diagram, and interpret percents greater than 100%.

WHY THIS MATTERS The arcade posts every result as a percent of the weekly goal. A percent compares a number to 100 — so 40% means 40 out of every 100. Some players land under the goal, and some blow past it: a score of 150% means one and a half times the whole goal. Your job is to read and model each result as a comparison to 100.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

3 out of 4 students bring lunch. What percent?

1. What times 4 makes 100? 25.
2. 3 × 25 = 75 and 4 × 25 = 100.
3. 75 out of 100 = 75%.

Answer: 75% bring lunch

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

What does the word percent mean?

(A) Per ten
(B) Per thousand
(C) Per hundred
(D) Per part

STEPS TO TRY
1. Look at the second half of the word: cent.
2. A century is 100 years; a cent is 1/100 of a dollar.
3. Percent = per 100.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

3 out of every 10 arcade players reach the bonus level. What percent is that?

(A) 3%

(B) 10%

(C) 13%

(D) 30%

STEPS TO TRY

1. A percent compares to 100, so rewrite the ratio out of 100.
2. What do you multiply 10 by to get 100?
3. Multiply BOTH terms by that same number.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A decimal grid has 100 squares and 65 are shaded. What percent is shaded?

(A) 6.5%

(B) 35%

(C) 65%

(D) 100%

STEPS TO TRY

1. How many squares are in the whole grid?
2. Each square stands for 1%.
3. Count the shaded squares.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which percent means MORE than one whole?

(A) 0.8%
(B) 80%
(C) 100%
(D) 120%

STEPS TO TRY

1. What percent stands for one whole?
2. Anything above that is more than the whole.
3. Compare each choice to 100.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

A friend says "a percent can never be more than 100, because you cannot have more than all of something." How would you answer using the mascot vote and the kittens?

SENTENCE STARTER A percent compares a number to ___, so ___% just means the amount is ___ than the whole it is compared to.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help